

財政學 HW5

1. (Hyman 5.2 改編)

The marginal benefit per gardener varies for five residents according to the following table:

Marginal Benefit per each gardener						
VOTER	1	2	3	4	5	6
Mike	350	300	200	150	100	50
Jan	100	80	60	40	20	0
Franklin	175	150	125	100	90	80
Susan	70	50	30	20	10	0
Megan	120	110	100	90	85	70

Marginal cost of hiring each gardener is \$400.

- (a) If each voter is assigned a tax share of \$80, what is the political equilibrium under majority rule?
- (b) Show that this equilibrium does not result in the efficient output of gardening services.
- (c) Show how the Lindahl equilibrium will differ from the political equilibrium under majority rule.

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(a) 多數決法則決策

每個人、每單位要負擔 80 → 80 是每個人享受服務的邊際成本多

請 1 人的邊際利益若……

大於 80，淨利益 > 0，投同意

小於 80，淨利益 < 0，投不同意

Q=5，有 3 人同意、2 人反對；Q=6，1 人同意、4 人反對

政治均衡 = 5 單位

(b) 效率均衡：公共財集體消費下，滿足 $\sum MB = MC$

政治決策出 Q=5

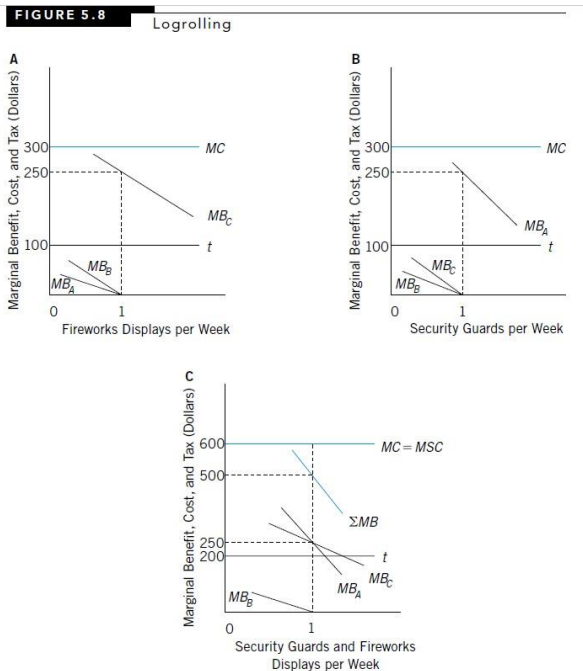
$100 + 20 + 90 + 10 + 85 < 400 \rightarrow$ 不效率

(c)

$\sum MB = MC$

林達爾均衡：

$Q=4$, $150+40+100+20+90 = 400$, 因此效率的量為 4 單位
均衡為 5 單位, 不效率



Logrolling can result in the passage of two issues together that could not pass if voted on separately.

2. (Hyman 5.3 改編)

The example of logrolling used in the text (Figure 5.8) assumes that the transactions costs of vote trading are zero. Suppose instead that voters A and C have to incur expenditures equal to \$80 per week to reach agreement on the vote-trading scheme. Show how this would prevent successful logrolling. Also show how logrolling would be impossible if the marginal benefit of the first security guard were only \$180 to voter A and transactions costs were zero.

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加入 A、C 交易成本 80 → 必須計入 A、C 各自的 MC

A 對保全的淨效用 = $250 - 100 - 80 = 70$

C 對煙火的淨效用 = $250 - 100 - 80 = 70$

淨效用(MB-MC)	煙火	保全	兩案均通過效用
A	-100	70	-30
B	-100	-100	-200
C	70	-100	-30

→都不通過效用較高，故不會選票交換

若 A 對保全的 MB 等於 180，無交易成本→保全淨效用為 80

A 對保全的淨效用 = $180 - 100 = 80$

因同意煙火的淨效用為-100→不接受交換的淨效用較高，故不會進行選票交換

淨效用(MB-MC)	煙火	保全	兩案均通過效用
A	-100	80	-20
B	-100	-100	-200
C	70	-100	-30

3. (Rosen 6.9)

Assume that the demand curve for milk is given by $Q = 100 - 10P$, where P is the price per gallon and Q is the quantity demanded per year. The supply curve is horizontal at the price of 2.

(a) Assuming that the market is competitive, what is the price per gallon of milk and the number of gallons sold?

(b) With the connivance of some politicians, the dairy farmers are able to form and maintain a cartel.(Such a cartel actually operates in the northeastern United States.)

What is the cartel price, and how many gallons of milk are purchased?[Hint: The marginal revenue curve (MR) is given by $MR = 10 - Q/5$. Also, remember that the supply curve shows the marginal cost associated with each level of output]

(c) What are the rent associated with the cartel?

(d) Suppose that in order to maintain the cartel, the dairy farmers simply give lump-sum campaign contributions to the relevant politicians. What is the maximum contribution they would be willing to make? What is the deadweight loss of the cartel?

(e) Suppose that instead of lump-sum contributions to politicians, the dairy farmers hire lobbyists and lawyers to make their case in Congress. How does this change your estimate of the deadweight loss associated with this rent-seeking activity?

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a. With the demand curve of $Q = 100 - 10P$ and a perfectly elastic supply curve at $P = 2$, the milk is sold at a price of \$2, and a quantity of 80 units is sold.

b. The marginal revenue curve associated with the inverse demand curve $P = 10 - (1/10)Q$ is $MR = 10 - (1/5)Q$, while the marginal cost curve is $MC = 2$. The cartel would

ideally produce a quantity where $MR=MC$, or $10-(1/5)Q=2$, or $Q=40$. The price associated with a cartel quantity of 40 units is $P=10-(1/10)*40$, or $P=6$.

c. The rent associated with the cartel is the product of the marginal profit per unit and the number of units produced. The marginal profit per unit of milk is \$4 (=\$6 price - \$2 marginal cost), while 40 units are produced. Thus, the rents equal \$160.

d. The most the cartel would be willing to contribute to politicians is the full economic rent of \$160. The cartel situation, the quantity of milk produced is too low from society's point of view. The deadweight loss triangle is computed using the difference between the cartel output and competitive output as the "base" of the triangle, and the difference between the cartel price and competitive price as the "height." Thus, the triangle is equal to $(1/2)*(80-40)*(\$6-\$2)=\$80$.

4. (Rosen 6.10 改編)

Consider a society with 3 people (John, Eleanor, and Abigali) who use majority rule to decide how much money to spend on public park.

There are 3 options for spending on a public park: H(high), M(median), L(low).

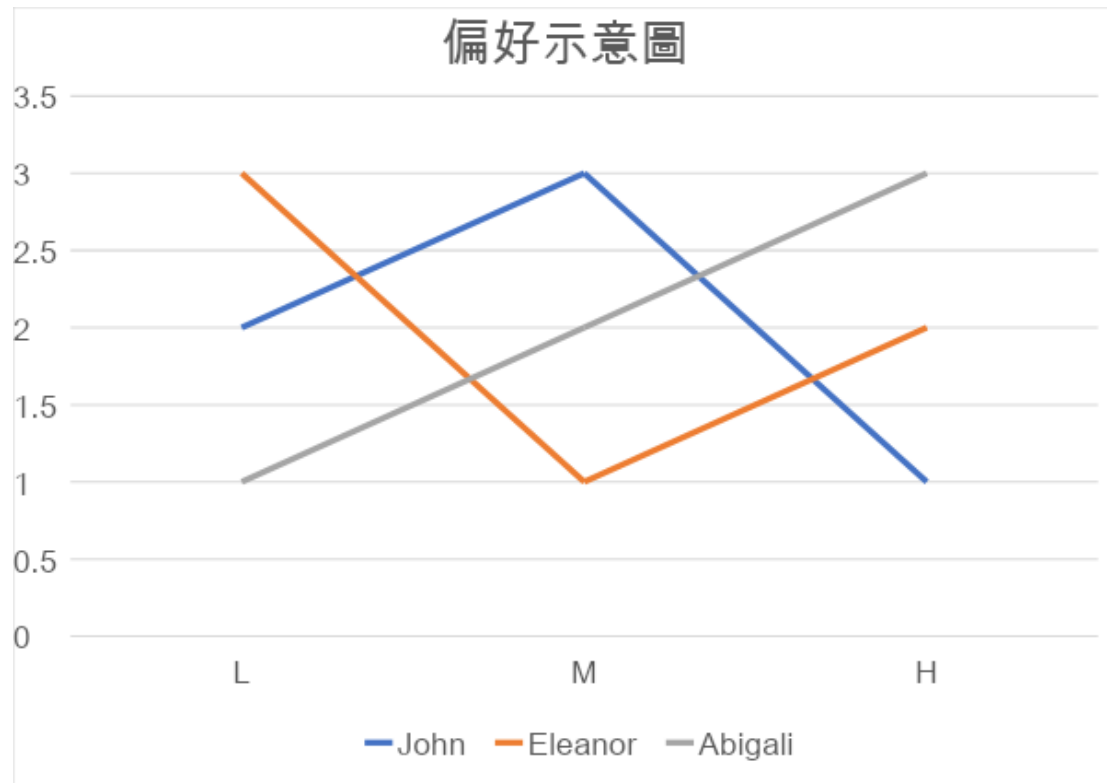
These individuals rank the 3 options in the following way:

Rank	John	Eleanor	Abigali
1	M	L	H
2	L	H	M
3	H	M	L

Consider all possible pairwise elections: M vs. H, H vs. L, L vs. M. What is the outcome of each election? Does it appear, in the case, that majority rule would lead to

a stable outcome on spending on the public park? If so, what is that choice? Would giving one person the ability to set the agenda affect the outcome? Explain

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組合	結果
M vs L	M
M vs H	H
L vs H	L

不會出現穩定的結果→循環會出現

議程的操縱有用→先排除會危及私心的案子

例如想讓 H 過→先決 M vs L，排除有威脅的 L 案

5. (Rosen 6.5)

In 2005, Kuwaiti women won the right to vote in parliamentary elections. Indeed,

women voters now outnumber men voters in Kuwait because women are automatically registered while men have to register on their own. One woman noted, “The Ministers of Parliament used to vote against us; now they are wooing us to vote for them” [Fattah, 2006]. What does this tell us about the validity of the predictions of the median voter theorem?

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When the policy changed to allow and encourage female voting, female voters became the majority, so it is sensible that the median voter is now a woman. Given this, the median voter theorem suggests that politicians should shift their positions to more closely align with the views of women.

6. (112-1期中考第4題)

Suppose there are five people A, B, C, D, and E in the society, and the following table shows their true values of a bridge construction plan:

people	A	B	C	D	E
$v^*(\$)$	30	40	50	60	70

Suppose the cost of the bridge is \$205.

Use what you learned from the Clarke-Groves mechanism to answer the following questions.

- Who are the decisive individuals?
- What are the payments of each individual?
- Is the total payment enough to cover the cost of the bridge?
- In general, what are the main problems of the Clarke-Groves mechanism?

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(a) C、D、E

(b) C: \$5、D: \$15、E: \$25

(c) 不夠

(d) 因為個人給付的金額與自身效益無關，有可能導致支付價金總額不夠支付公共財成本。

7. 市場內有 X 財貨與 Y 財貨，且已知兩財貨的需求函數分別為 $X(P_X, M)$ 與 $Y(P_Y, M)$ ，M 是所得，故可以知道預算限制式為 $P_X \cdot X + P_Y \cdot Y = M$ 。

X 財貨份額為 $(P_X \cdot X)/M$ 。請以數學推導以下兩命題：[提示：整理一階微分的結果]

- 當 X 財貨的價格彈性小於 1 時，X 財貨份額隨 X 財貨價格增加而增加
- 當 X 財貨的預算彈性大於 1 時，X 財貨份額隨所得增加而增加

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$$(1) \frac{\partial \frac{P_X \cdot X}{M}}{\partial P_X} = \frac{X}{M} + P_X \cdot \frac{\partial X}{\partial P_X} \cdot \frac{1}{M} = \frac{X}{M} \left[1 - \left(-\frac{\partial X}{\partial P_X} \cdot \frac{P_X}{X} \right) \right] > 0$$

$$(2) \frac{\partial \frac{P_X \cdot X}{M}}{\partial M} = \frac{P_X}{M} \cdot \frac{\partial X}{\partial M} - P_X \cdot X \cdot \frac{1}{M^2} = \frac{P_X \cdot X}{M^2} \left[\left(\frac{\partial X}{\partial M} \cdot \frac{M}{X} \right) - 1 \right] > 0$$